THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · EASY LEVEL CHAPTER TEST SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The apex angle of an isosceles triangle is 100°. Each base angle is: 1

(a) 30° **(b)** 40° **(c)** 50° **(d)** 60°

2. The perimeter of a triangle with sides 10 cm, 12 cm and 15 cm (in cm) is: 1

(a) 36 **(b)** 38 **(c)** 39 **(d)** 37

3. Two triangles with two angles and the side between them equal are congruent by: 1

(a) AAA **(b)** SSS **(c)** ASA **(d)** SAS

4. In a right-angled triangle one acute angle is 52°. The other acute angle is: 1

(a) 58° **(b)** 38° **(c)** 48° **(d)** 28°

5. Can a triangle have sides 4 cm, 4 cm and 8 cm? 1

(a) Yes **(b)** Only if it is right-angled

(c) No **(d)** Only if it is isosceles

6. Two angles of a triangle are 64° and 46°. The third angle is: 1

(a) 80° **(b)** 90° **(c)** 60° **(d)** 70°

7. Each angle of an equilateral triangle measures: 1

(a) 60° **(b)** 30° **(c)** 90° **(d)** 45°

8. In a triangle, the two interior opposite angles are 28° and 81°. The exterior angle is: 1

(a) 109° **(b)** 119° **(c)** 129° **(d)** 99°

9. The apex angle of an isosceles triangle is 120°. Each base angle is: 1

(a) 30° **(b)** 50° **(c)** 40° **(d)** 20°

10. The perimeter of a triangle with sides 8 cm, 8 cm and 8 cm (in cm) is: 1

(a) 26 **(b)** 25 **(c)** 23 **(d)** 24

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$, $\angle A = 72^\circ$ and $\angle B = 38^\circ$. Side BC is extended to D. Find the exterior angle $\angle ACD$. 2

12. Can a triangle be formed with sides 4 cm, 6 cm and 10 cm? Give a reason. 2

13. Two angles of a triangle are 55° and 55°. Find the third angle and name the triangle by its angles. 2

14. In $\triangle ABC$, $\angle A = 55^\circ$ and $\angle B = 50^\circ$. Side BC is extended to D. Find the exterior angle $\angle ACD$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The angles of a triangle are in the ratio $2 : 3 : 5$. Find the angles. 3

16. The apex angle of an isosceles triangle is 20°. Find the base angles. 3

17. An exterior angle of a triangle is 150° and one interior opposite angle is 70°. Find the other interior opposite angle and the adjacent interior angle. 3

OR

The angles of a triangle are in the ratio $1 : 3 : 5$. Find the angles. 3

18. A base angle of an isosceles triangle is 35°. Find the apex angle. 3

OR

An exterior angle of a triangle is 125° and one interior opposite angle is 52°. Find the other interior opposite angle and the adjacent interior angle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A triangle has sides 10 cm, 10 cm and 15 cm. Check that it exists, name it by its sides and find its perimeter. 5

OR

In $\triangle ABC$, $\angle A = 60^\circ$ and $\angle B = 60^\circ$. Find $\angle C$, classify the triangle by its angles and find the exterior angle at C. 5

20. A triangle has sides 4 cm, 7 cm and 9 cm. Check that it exists, name it by its sides and find its perimeter. 5

OR

In $\triangle ABC$, $\angle A = 25^\circ$ and $\angle B = 45^\circ$. Find $\angle C$, classify the triangle by its angles and find the exterior angle at C. 5

END OF QUESTION PAPER — Total: 40 Marks

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